

Test Form D

Name _____ Date _____

Chapter 11

Class _____ Section _____

1. Find a unit vector in the direction of $\mathbf{v}$ if $\mathbf{v}$ is the vector from $P(2, 1, -3)$ to $Q(-1, 0, 4)$.

2. Calculate $\mathbf{i} \times \mathbf{k}$.

In problems 3 through 8, let $\mathbf{u} = -\mathbf{i} + 2\mathbf{j} - \mathbf{k}$, $\mathbf{v} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{w} = 3\mathbf{i} - \mathbf{k}$.

3. Calculate $\mathbf{u} \cdot \mathbf{v}$.

4. Calculate $\mathbf{u} \times \mathbf{v}$.

5. Calculate $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$.

6. Calculate $\text{proj}_{\mathbf{v}}\mathbf{u}$.

7. Calculate θ where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$.

8. Calculate a vector perpendicular to both $\mathbf{u}$ and $\mathbf{w}$.

9. Let $L_1 : x = 2 - 4t, y = -1 + 5t, z = 7t$ and $L_2 : x = 2 + t, y = 5 - 2t, z = 1 + 2t$. Show that the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ parallel to lines L_1 and L_2 , respectively, are orthogonal.

10. Find parametric equations for the line through the point $(4, -1, 3)$ and parallel to the line given by $\frac{x}{1} = \frac{y-2}{2} = \frac{z+1}{5}$.

11. Write an equation of the plane that is determined by the lines given by $\frac{x-6}{-6} = \frac{y+1}{2} = \frac{z}{-3}$ and $\frac{x-4}{2} = \frac{y-3}{1} = \frac{z+1}{1}$.

12. Sketch the plane given by $3x + 4y + 2z = 12$.

13. Find the numbers x and y such that the point $(x, y, 1)$ lies on the line passing through the points $(2, 5, 7)$ and $(0, 3, 2)$.

14. Write the equation of the sphere, $4x^2 + 4y^2 + 4z^2 - 4x + 16y - 8z + 9 = 0$, in standard form and find the center and radius.

15. Identify and sketch the quadric surface given by $\frac{x^2}{1} + \frac{y^2}{4} - z = 0$.

16. Sketch the surface represented by the equation $y = x^2$.
17. Find the equation of the surface of revolution if the generating curve, $2x + 3z = 1$, is revolved about the x -axis.
18. Convert the point $(3, -3, 7)$ from rectangular to cylindrical coordinates.
19. Find an equation in rectangular coordinates for the spherical coordinate equation $\rho = 9 \csc \phi \csc \theta$ and identify the surface.
20. Find the area of the parallelogram having vectors $\mathbf{u} = \mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$ and $\mathbf{v} = -5\mathbf{i} + 2\mathbf{k}$ as adjacent sides.